Exposé: diploma thesis Porsche Motorsport EMA3

1. Title

Development of an software analysis tool for MATLAB/Simulink models and their generated C Code, providing a graphical signal dependency overview and computation data that can be used to optimize runtime and reduce cross-core data transitions on a Motorsport Vehicle Control Unit.

2. Motivation/Problem Statement

At Porsche Motorsport, a large part of the software deployed on the Formula E and LMDH vehicles is C code generated from MATLAB/Simulink models. These models are constantly updated and optimized by Porsche engineers to improve the vehicle's performance. If a certain function in one of these models needs to be changed, the engineer currently has to navigate through the model to track signal dependencies and understand the signal composition for all necessary inputs of the function to be improved. Tracing a signal through the model this way is time consuming and generates a cognitive overhead for these engineers. The engineers need a quick overview of individual signal composition and the origin of computation.

3. Objective

The objective of this thesis is to develop a software analysis tool for MATLAB/Simulink models that accepts an SLX model or the generated Code and Artifacts from MATLAB Embedded Coder as input. By using a user-selected signal (which can be an output of a block or a signal in the model), the tool will perform backward tracing of this signal through the model. The data gained from this analysis will be used to create a structural calculation tree that shows all blocks/values/signals/inputs that contribute to the signal. The result will be a readable map that instantly shows all relationships of the selected signal, containing additional information such as the origin of computation for each block. This gives engineers a fast way to understand, review, and document signal dependencies without manually navigating through the model.

4. Methodology/Procedure

- Analysis of generated C code and Embedded Coder artifacts to establish the relationship between Simulink blocks and generated code functions
- Architectural design of the analysis tool: specification of the parsing system, data structures for backward signal tracing, integration of the database for block metadata, and interface between model input and visualization output
- Edge case identification and and handling for complex Simulink models, including Stateflow, Bussystems, custom blocks, Lookup tables, and other non-standard blocks
- Codebase development using AI-assisted coding tools, with the student responsible for task decomposition, code review, and validation against defined requirements
- Verification of tool output against manually traced signals from existing Simulink models

5. Optional Extension / Benefits

Goal: Reduce cross-core communication overhead in the multicore Motorsport Vehicle Control Unit and enable more efficient task scheduling. As an example, all C functions belonging to a critical telemetry data channel's calculation tree could be placed on the same core, reducing the number of cross-core signal transfers per execution round and the resulting latency. This is enabled by the signal dependency data extracted during backward tracing, combined with additional information from the generated C code and Embedded Coder artifacts.

Based on this, the following strategies will be evaluated:

- Creating Clusters of signals and their computing blocks by shared dependency, placing tightly coupled groups on the same core to minimize cross-core transitions.
- Identifying time relevant signal chains with the highest cross-core transfer frequency, and prioritizing these for relocation.
- Analyzing Embedded Coder artifacts to identify configured rate-transition mechanisms (e.g., buffering vs. semaphores) and evaluating their runtime cost based on the execution code.

The tool provides placement *recommendations*; no automatic re-placement or code modification is performed (see Scope)

6. Scope

- The focus is on the development of the analysis/mapping tool
- multicore optimization suggestions are supplementary/optional
- there is no automatic re-placement

7. Support

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